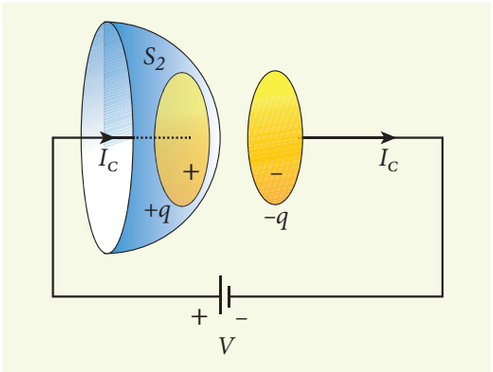


Figure 5.5 Applying Ampere's circuital law - loop enclosing surface S2

$$\oint_{\text{enclosing } S_2} \vec{B}\cdot d\vec{l} = 0 \qquad (5.5)$$

The right hand side of equation is zero because the surface S_2 no where touches the wire carrying conduction current and further, there is no current in between the plates of the capacitor (there is a discontinuity). So the magnetic field at a point P is zero. Hence there is an inconsistency between equation (5.4) and equation (5.5). J. C.Maxwell resolved this inconsistency as follows:

Due to external source (battery or cell), the capacitor gets charged up because of current flowing through the capacitor. This produces an increasing electric field between the capacitor plates. So, there must be a current associated with the changing electric field in between the capacitor plates. In other words, the time varying electric flux (or time varying electric field) existing between the plates of the capacitor also produces a current known as displacement current.

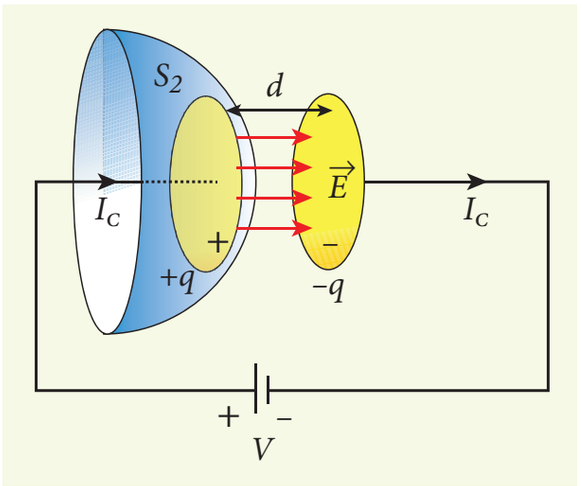


Figure 5.6 Applying Gauss's law between the plates of the capacitor

From Gauss's law (refer Unit 1), the electric flux between the plates of the capacitor (Figure 5.6) is

$$\Phi_E = \iint \vec{E}\cdot d\vec{A} = EA = \frac{q}{\varepsilon_0}$$

where A is the area of the plates of capacitor.

The change in electric flux is

$$\frac{d\Phi_E}{dt} = \frac{1}{\varepsilon_\circ}\frac{dq}{dt} \Rightarrow \frac{dq}{dt} = I_d = \varepsilon_\circ \frac{d\Phi_E}{dt}$$

where I_d is known as displacement current. The **displacement current can be defined as the current which comes into play in the region in which the electric field and the electric flux are changing with time**. In other words, whenever the change in electric field takes place, displacement current is produced. Maxwell modified Ampere's law as

$$\oint \vec{B}\cdot d\vec{S} = \mu_\circ I = \mu_\circ \left(I_C + I_d\right) \qquad (5.6)$$

Note

Displacement current

The name stuck because Maxwell named it. The word displacement is poorly chosen because nothing is being displaced here.

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